EV VEHICLE TRACKER | SIM800L EDITION

Prototype Programming & Test Guide

GPS tracking | Engine ON/OFF monitor | Battery-removal alarm + SMS | Remote immobiliser

Item	Value
Revision	3.0 (25 Oct 2026). SIM800L version of the Rev 2.0 (SIM7600) guide
Firmware	**EV_Tracker_v3_SIM800L.ino** (main), **SIM800L_AT_Test.ino** (modem bring-up), **CustomFirebaseFS.h** (library setting)
Schematic	**EVT-SCH-03**, A3, on the next page and supplied as a separate file
Network	SIM800L, 2G GSM/GPRS. **Airtel** SIM recommended (Vi / BSNL also work). **Jio will not work** (no 2G).
Alerts	Firebase Realtime Database for the app, plus an SMS with a Google Maps link to the owner
Target	ESP32 DevKit V1 on a 12 V bench supply; Wi-Fi bench mode available (NET_MODE 0)

What changes with the SIM800L

Area	SIM800L design (this guide)	Why
Modem power	Fed straight from the 18650 cell (VBAT_LI, 3.7-4.2 V) with 1000 uF + 100 nF at the pins.	The SIM800L needs 3.4-4.4 V and 2 A bursts. 5 V destroys it; weak supplies make it reset.
UART level	ESP32 TX to SIM RXD through 1k / 3k3 divider (about 2.5 V).	SIM800L I/O is 2.8 V logic.
Reset	SIM RST on GPIO4, open-drain. Firmware resets a hung modem.	SIM800L can lock up after a brown-out.
Data rate	Upload every 30 s, command check every 15 s (Wi-Fi mode: 5 s).	GPRS is slow; each HTTPS call takes 2-8 s.
Commands	App command is polled (no stream).	Holding a stream open over GPRS is unreliable.
Alarm	SMS to the owner with a maps link, plus the Firebase alert.	SMS works even when GPRS data is down.
Library	One extra file, CustomFirebaseFS.h, copied into the Firebase library.	Turns on GSM support inside the library.

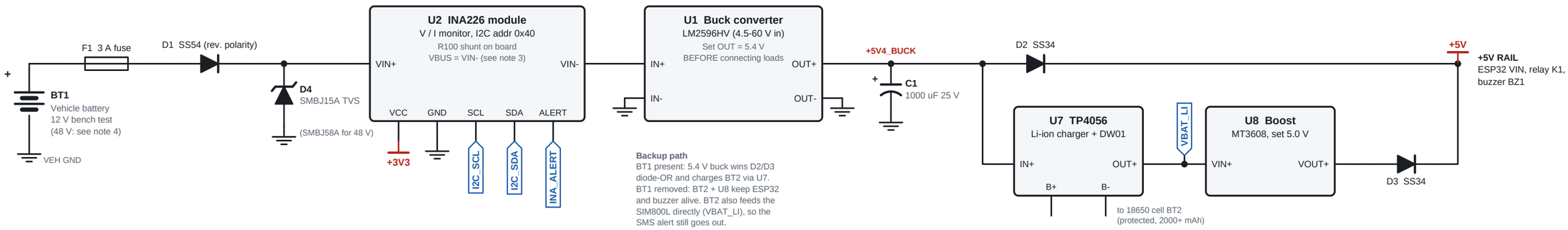
Safety first. Bench-test with a current-limited 12 V supply (1 A limit) and a 12 V lamp standing in for the ignition circuit. Connect to a real vehicle only after every test in Section 7 passes, and never test the immobiliser on a moving vehicle.

2G only. India's 2G networks are being phased out. This build is right for a prototype or college demo; for a long-life product, move to a 4G module (the Rev 2.0 SIM7600 design).

EV VEHICLE TRACKER - PROTOTYPE SCHEMATIC (SIM800L / 2G)

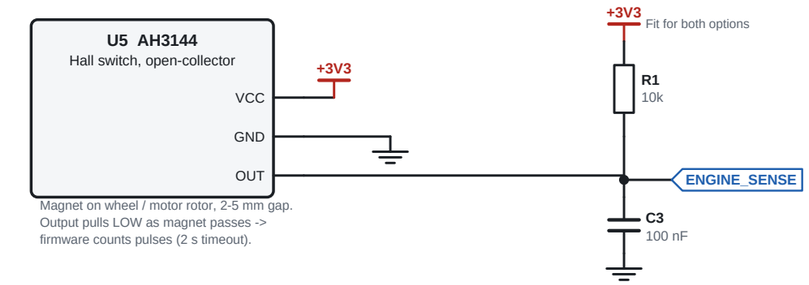
Matches firmware EV_Tracker_v3_SIM800L.ino (pin map frozen). Net labels with the same name are connected.

A. POWER, PROTECTION, BACKUP & BATTERY SENSE

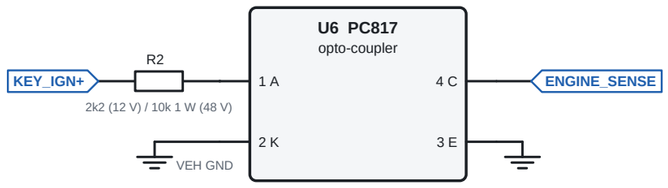


B. ENGINE SENSE (fit ONE option)

OPTION A - Hall sensor (ENGINE_SENSE_MODE 0)

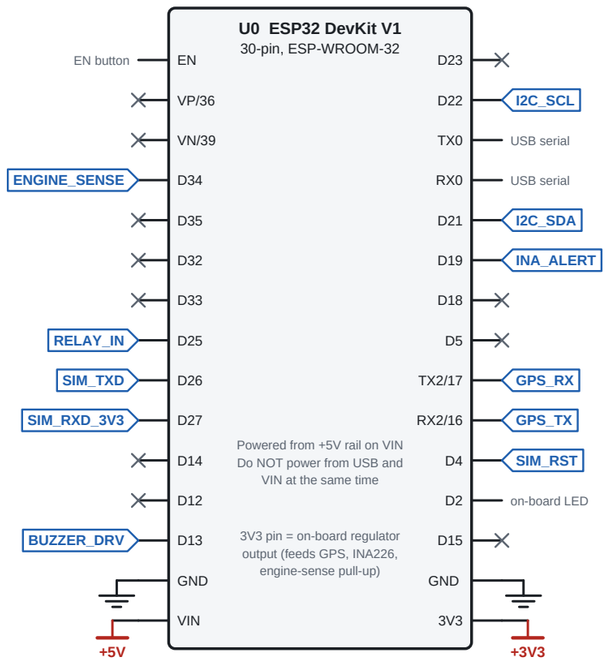


OPTION B - Ignition sense (ENGINE_SENSE_MODE 1)

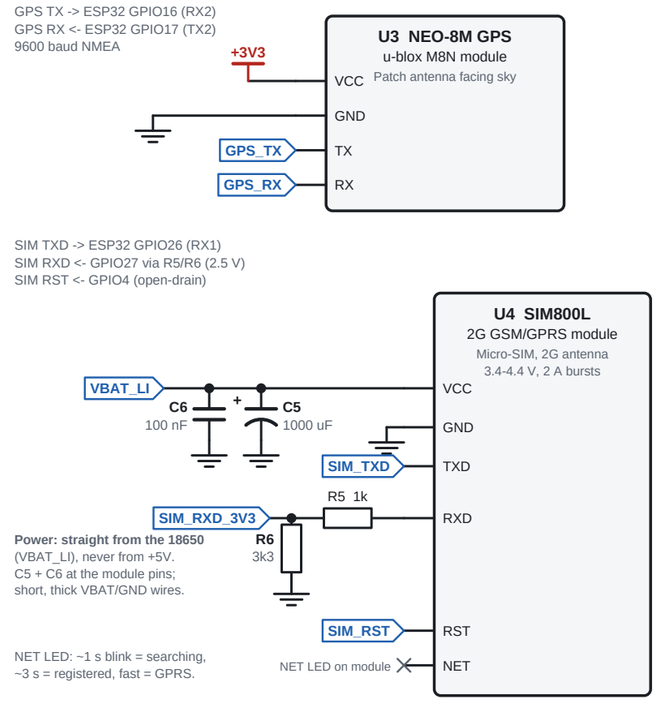


Ignition ON -> LED on -> transistor pulls ENGINE_SENSE LOW.
Isolates vehicle wiring from the ESP32.

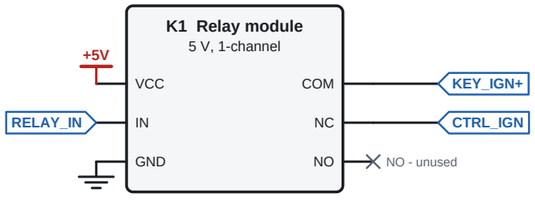
C. MICROCONTROLLER



D. GPS + 2G GSM MODULE (SIM800L)

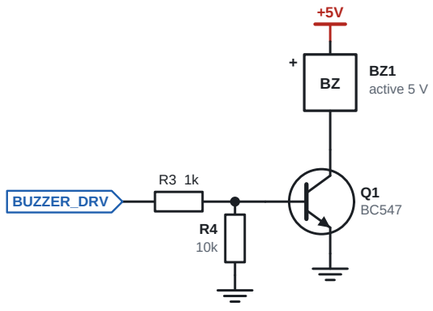


E. IMMOBILISER RELAY (fail-safe)



KEY_IGN+ = key-switch output. CTRL_IGN = motor-controller ignition / key input.
Break the key line and pass it through COM-NC. Low-current signal only, never the traction line.
De-energised (boot, power loss, fault) -> NC closed -> vehicle RUNS.
Energised (app sends "OFF", speed <= 5 km/h) -> NC opens -> ignition CUT.
Set RELAY_ACTIVE_LOW in firmware to match your module.

F. ALARM BUZZER



GPIO13 HIGH -> Q1 on -> buzzer sounds.
R4 holds Q1 off during boot.

G. NOTES

1. All GND symbols are one common ground: vehicle battery -, buck, TP4056, boost, ESP32, modules.
2. GPIO34 is input-only with no internal pull-up: R1 (10k to +3V3) is mandatory.
3. INA226: power VCC from +3V3 (keeps I2C pull-ups at 3.3 V). VBUS max 36 V. On most boards VBUS is tied to VIN- on the PCB; if yours has a separate VBUS pin, connect it to VIN-.
4. 48 V vehicles: U1 must be a 60 V+ part (LM2596HV / XL7015), D4 = SMBJ58A, and the INA226 VBUS must see < 36 V (divider into VBUS + set BATT_DIVIDER_RATIO) or use an INA228 (85 V).
5. Build order: power -> backup (BT2) -> ESP32 -> SIM800L -> GPS -> INA226 -> engine sense -> buzzer -> relay. SIM800L works only on 2G (Airtel / Vi / BSNL SIM; Jio will NOT work).
6. Pins kept free of ESP32 boot-strapping pins (0, 2*, 5, 12, 15). *GPIO2 only drives on-board LED.
7. SIM800L logic is 2.8 V: R5/R6 drop the ESP32 TX to ~2.5 V. Never drive RST HIGH from the ESP32.

EV Vehicle Tracker - SIM800L Prototype	
Doc: EVT-SCH-03 Rev 3.0 Sheet 1 of 1	Date: 25 Oct 2026
Firmware: EV_Tracker_v3_SIM800L.ino Board: ESP32 DevKit V1	Size A3

1. Frozen pin map

These pins match **EV_Tracker_v3_SIM800L.ino** and schematic **EVT-SCH-03**. Change all three together or not at all.

ESP32 pin	Net	Connects to	Notes
GPIO16 (RX2)	GPS_TX	NEO-8M TX	UART2, 9600 baud
GPIO17 (TX2)	GPS_RX	NEO-8M RX	
GPIO26 (RX1)	SIM_TXD	SIM800L TXD	UART1, 57600 baud (9600 fallback)
GPIO27 (TX1)	SIM_RXD_3V3	R5 1k to SIM800L RXD, R6 3k3 to GND	Divider brings 3.3 V down to ~2.5 V
GPIO4	SIM_RST	SIM800L RST	Open-drain: firmware pulls LOW or releases; never drives HIGH
GPIO21 / GPIO22	I2C_SDA / SCL	INA226 SDA / SCL	INA226 address 0x40, powered from 3V3
GPIO19	INA_ALERT	INA226 ALERT	Optional, unused
GPIO34	ENGINE_SENSE	AH3144 OUT or PC817 collector	Input-only; R1 10k pull-up to 3V3 is mandatory
GPIO25	RELAY_IN	Relay module IN	Polarity set by RELAY_ACTIVE_LOW
GPIO13	BUZZER_DRV	R3 1k to BC547 base	Buzzer on the 5 V rail
GPIO2	-	On-board blue LED	Solid = Firebase ready, blinking = not connected
VIN / 3V3	+5V / +3V3	5 V rail / GPS, INA226, R1	SIM800L is NOT on either rail: it uses VBAT_LI

2. Prototype bill of materials

Ref	Part	Qty	Purpose
U0	ESP32 DevKit V1 (30-pin)	1	Main controller
U4	**NEW** SIM800L module (red mini board) + 2G antenna (IPEX pigtail preferred over the spring)	1	2G GPRS data + SMS
SIM	**NEW** Airtel prepaid micro-SIM (or nano-SIM with adapter), PIN disabled	1	Network
U3	u-blox NEO-8M / M8N GPS module + patch antenna	1	Location, speed
U2	INA226 module (R100 shunt)	1	Battery voltage / tracker current
U1	LM2596HV buck module (XL7015 for 48-72 V)	1	Vehicle battery to 5.4 V
U7 / U8	TP4056 with DW01 protection, MT3608 boost	1 each	Charge BT2; backup 5 V rail
BT2	18650 Li-ion, **2500 mAh+, high-drain** (e.g. 10 A rated) + holder	1	Powers SIM800L and backup
C5 / C6	**NEW** 1000 uF 10 V low-ESR electrolytic, 100 nF ceramic	1 each	SIM800L burst current
R5 / R6	**NEW** 1k, 3k3	1 each	UART level divider
U5 / U6	AH3144 + magnet (Option A) or PC817 (Option B)	1	Engine sense
K1	5 V 1-channel relay module	1	Immobiliser
BZ1 / Q1	Active 5 V buzzer, BC547	1 each	Local alarm
F1, D1-D4	3 A fuse; SS54, 2x SS34, SMBJ15A	-	Protection, diode-OR
C1, C3, R1-R4	1000 uF 25 V; 100 nF; 10k, 2k2, 1k, 10k	-	Filtering, pull-ups, drivers
-	Perfboard, 20 AWG wire for VBAT/GND, IP65 box, 12 V 2 A bench supply, multimeter, 12 V lamp	-	Build and test

3. Build order

Build and test one stage at a time; each stage has a matching test in Section 7. The SIM800L comes early because the backup cell it runs on must be working first.

Stage	Build	Test
1 Power	F1, D1, D4, INA226 shunt path, buck U1, C1, D2. **Set U1 to 5.4 V with no load.**	T1
2 Backup	TP4056 from +5V4_BUCK, 18650, MT3608 **set to 5.0 V before** fitting D3. Charge the cell fully.	T2
3 ESP32	VIN to +5V rail, GND common. Flash over USB with the 12 V supply OFF.	T5
4 SIM800L	VCC from VBAT_LI with C5 + C6 soldered at the module pins; GND; TXD to GPIO26; RXD via R5/R6 from GPIO27; RST to GPIO4. Antenna fitted, SIM inserted.	T3, T4, T6
5 GPS	NEO-8M on 3V3, TX to GPIO16, RX to GPIO17.	T7
6 INA226	VCC 3V3, SDA 21, SCL 22. Shunt between D4 node and buck input.	T8
7 Engine sense	R1 + C3, then Option A (AH3144) or B (PC817). Set ENGINE_SENSE_MODE.	T9
8 Buzzer, relay	BC547 driver + buzzer; relay COM-NC switching the 12 V test lamp.	T10-T14

Never connect the SIM800L to 5 V. Its absolute maximum is about 4.5 V. Measure VBAT_LI (3.7-4.2 V) before plugging the module in. Use short, thick wires (20 AWG) for VBAT and GND: thin jumper wires drop enough voltage during 2 A bursts to make the modem reset.

4. Programming the ESP32

4.1 Install the tools

1. Install **Arduino IDE 2.x** and the USB driver for your board (**CP210x** or **CH340**).
2. File > Preferences > Additional boards manager URLs:
   `https://espressif.github.io/arduino-esp32/package_esp32_index.json`
3. Boards Manager: install **esp32 by Espressif Systems** (use **2.0.17** if the Firebase library fails to compile on 3.x).
4. Library Manager: install **TinyGPSPlus** (Mikal Hart), **TinyGSM** (Volodymyr Shymanskyy), **Firebase Arduino Client Library for ESP8266 and ESP32** (Mobizt, v4.4.x) and **INA226** (Rob Tillaart).

4.2 One-time library step (GPRS will not work without it)

Copy the supplied **CustomFirebaseFS.h** into the Firebase library's **src** folder. It contains one line:

```
// File: Documents/Arduino/libraries/
//       Firebase_Arduino_Client_Library_for_ESP8266_and_ESP32/src/CustomFirebaseFS.h
#define TINY_GSM_MODEM_SIM800
```

The library only compiles its GSM support when this macro is visible inside the library itself; defining it in the sketch alone is not enough. Re-copy the file after updating the library.

4.3 Board settings and upload

Setting	Value
Board	ESP32 Dev Module
Partition Scheme	Huge APP (3MB No OTA/1MB SPIFFS)
Upload Speed	921600 (115200 if uploads fail)
Serial Monitor	115200 baud, line ending: Newline

1. First flash **SIM800L_AT_Test/SIM800L_AT_Test.ino** and run tests T3-T4 before the main firmware.
2. Then open **EV_Tracker_v3_SIM800L/EV_Tracker_v3_SIM800L.ino**, edit USER CONFIG (table below), Verify, and Upload with the 12 V supply off.
3. If upload says *Failed to connect*, hold **BOOT** when "Connecting..." appears.

Setting	Set to
NET_MODE	1 = SIM800L GPRS (vehicle). 0 = Wi-Fi (bench); SMS still goes through the SIM800L.
GPRS_APN	Airtel: **airtelgprs.com**. Vi: www. BSNL: bsnlnet.
OWNER_PHONE	Your number with country code, e.g. +9198xxxxxxxx
SMS_ENABLED	true. Set false to stop SMS during repeated bench tests.
SIM_PIN	Leave empty; disable the PIN in a phone instead.
API_KEY, DATABASE_URL, USER_EMAIL/PASSWORD	From Firebase (Section 6). URL without https://
ENGINE_SENSE_MODE / RELAY_ACTIVE_LOW	As in schematic options; confirm in T9 and T11.
SIM_BAUD (pin map block)	57600. Change to 9600 if AT replies look garbled.

5. Preparing the SIM card

1. Buy an **Airtel prepaid** SIM (any new SIM works on 2G). Ask for a micro-SIM, or use a nano-to-micro adapter.
2. Put it in a phone first: activate it, **turn off the SIM PIN lock**, and send one SMS to check SMS works.
3. In the phone, set the network to 2G only and open a web page: this proves 2G data works at your location.
4. Recharge a plan that includes SMS. Data use at 30 s uploads is roughly **150-300 MB per month**, so any data plan is enough.
5. Insert the SIM into the SIM800L with the cut corner as printed on the board, with power OFF.

6. Firebase setup

1. Firebase console: Add project. Build > **Realtime Database** (asia-southeast1, locked mode).
2. Build > **Authentication**: enable **Email/Password**; add a **device** user and an **owner** user; copy both UIDs.
3. Copy the **Web API Key** and database URL into the firmware. Paste the rules below with your UIDs and Publish.
4. Create `EV_TRACKER_01/commands/engineControl = "ON"` by hand once.

```
{ "rules": { "EV_TRACKER_01": {
    ".read":  "auth != null && (auth.uid == 'DEVICE_UID' || auth.uid == 'OWNER_UID')",
    ".write": "auth != null && auth.uid == 'DEVICE_UID'",
    "commands": { "engineControl": {
        ".write": "auth != null && auth.uid == 'OWNER_UID'",
        ".validate": "newData.isString() && newData.val().matches(/^(ON|OFF|CLEAR)$/)" } } } } }
```

Path under /EV_TRACKER_01	Type	Meaning
gps/lat, lng, speedKmph, sats, fix	number / bool	Position and speed (when fix valid), satellites, fix
engine/status, immobilised, immobilisePending	bool	Engine on; relay state; cut queued while moving
battery/voltage, current_mA, percent, present, alarm	number / bool	INA226 readings and removal alarm
alerts/{pushId}	object	type, voltage, lat, lng, time per removal event
device/net, device/csq, device/uptimeS, lastUpdated	mixed	GPRS/WIFI, 2G signal (10-31 good), uptime, server time (every 30 s)
commands/engineControl	string	**Written by the app:** OFF, ON or CLEAR. Device checks every 15 s.

7. Bench test procedure

Main-firmware menu: **h** help, **s** status, **m** modem info, **t** test SMS, **p** AT pass-through (**~** exits), **k** check command now, **u** upload now, **r** relay, **b** beep, **c** clear alarm, **g** raw GPS, **i** I2C scan. Status line every 2 s:

```
GPS fix:Y sats:8 lat:12.924512 lng:80.127345 spd:0.0 | ENG:OFF (0.0 p/s) |
  BATT:12.41V 182mA 87% OK | RELAY:RUN | CSQ:18 FB:Y
```

#	Test	How	Pass when	Result
T1	Power rail	12 V supply, 1 A limit, no modules. Measure U1 out, +5V rail; briefly reverse the leads.	U1 = 5.4 V, rail 4.9-5.1 V; reversed = no current (D1).	
T2	Backup + VBAT_LI	Charge the 18650. Measure VBAT_LI and U8 output. Switch 12 V off: rail must stay up.	VBAT_LI 3.7-4.2 V, U8 5.0 V, +5V rail stays above 4.5 V on backup.	
T3	SIM800L AT	Flash SIM800L_AT_Test.ino. Watch Serial and the NET LED.	AT OK; +CBC 3700-4200 mV; +CPIN: READY; CSQ 10-31; CREG 0,1 or 0,5; SAPBR gives an IP; NET LED blinks every ~3 s.	
T4	SMS	In the AT test type **sms** (set PHONE first).	+CMGS reply and the SMS arrives on your phone.	
T5	Main fw boot	Flash EV_Tracker_v3_SIM800L.ino with NET_MODE 1. Open Serial.	Banner, "[OK] Modem", "[OK] Network: airtel CSQ n", "[OK] GPRS up, IP ...".	
T6	GPRS + Firebase	Wait up to 60 s for the token, then watch uploads; open the Firebase console.	FB:Y, "[OK] Uploaded (2000-8000 ms)" every 30 s; data under /EV_TRACKER_01, device/net = GPRS.	
T7	GPS	g indoors (NMEA streams), then open sky.	Fix within ~1-5 min, 6+ sats, position within ~5 m of a phone map.	
T8	INA226	i, then compare BATT with a multimeter; sweep 11-13 V.	0x40 found; within 0.1 V; tracks the sweep.	
T9	Engine sense	A: wave magnet past AH3144, stop. B: 12 V on KEY_IGN+, remove.	ENG:ON while pulsing / with 12 V; OFF within 2 s after.	
T10	Buzzer	b	1 s beep; silent at boot.	
T11	Relay	Meter on COM-NC; r twice.	IMMOBILISED = COM-NC open; RUN = closed. Wrong way: flip RELAY_ACTIVE_LOW.	
T12	Remote command	Lamp on COM-NC. Set commands/engineControl to OFF, then ON (or press k).	Within 15 s: ">> Command received: OFF", lamp off; ON restores it.	
T13	Fail-safe	Relay IMMOBILISED, then cut all tracker power (pull BT2 too).	Relay drops, lamp on (vehicle would run).	
T14	Battery removal	BT2 charged, SMS_ENABLED true. Switch the 12 V supply off.	Buzzer within ~0.3 s; ESP32 + SIM800L stay up on BT2; SMS with maps link within ~30 s; alerts/ entry in Firebase.	
T15	Battery restore	Switch 12 V back on, then send CLEAR (or c).	"battery reconnected" SMS; present = true; buzzer stops on CLEAR.	
T16	Signal loss	Unscrew the antenna (or wrap the module in foil) for 2 min, then restore.	Uploads fail with errors, local alarm still works; recovers by itself within ~1-2 min.	
T17	Brown-out	Watch Serial during 10 uploads and one SMS.	No "SIM800L not responding" resets and no ESP32 reboots.	
T18	Soak	2 h run, GPS near a window.	uptimeS keeps rising; upload gaps under 90 s.	

Speed guard (immobiliser refused while moving): test as a passenger in a car with the tracker on its own battery and the relay switching only the test lamp. Send OFF while moving: "immobilise queued"; it engages only after the car stops.

8. Troubleshooting

Symptom	Likely cause	Fix
No reply to AT	TX/RX swapped, no power, or baud	SIM TXD to GPIO26; check VBAT_LI; try SIM_BAUD 9600
NET LED off, or modem keeps restarting	Supply dips during 2 A bursts	Charged high-drain 18650, C5 at the pins, thick short wires
+CPIN: NOT READY / NOT INSERTED	SIM orientation, PIN on, bad adapter	Re-seat SIM, disable PIN in a phone, try a real micro-SIM
CSQ 99 or CREG 0,2 forever	No 2G signal, Jio SIM, or no antenna	Use Airtel/Vi; fit antenna; test near a window
GPRS fails (SAPBR or gprsConnect)	Wrong APN or no data pack	airtelgprs.com; check the balance in the phone
Compiles but Firebase never uses GPRS	CustomFirebaseFS.h missing in the library	Copy it to the library src folder (Section 4.2), rebuild
Firebase SSL / timeout errors	Weak signal or slow first connection	Wait for CSQ 10+; first token can take 60 s; keep uploads at 30 s
SMS not received	No SMS balance, wrong number format	Recharge SMS; OWNER_PHONE as +91XXXXXXXXXX
INA226 not found / ENG flickers / relay wrong way	See Rev 2.0 causes	Check 3V3 + SDA/SCL; fit R1; flip RELAY_ACTIVE_LOW
ESP32 resets when the relay clicks	5 V rail dip	Check C1, buck rating; unplug USB when running on VIN

9. Known limits

- **2G sunset:** when Airtel/Vi switch off 2G in your area, the SIM800L stops working. Plan a move to a 4G module for real use.
- **Latency:** app commands take up to 15 s; positions update every 30 s. Shorter intervals use more data and stress the cell.
- **Battery percent** is a linear estimate; replace with a lookup table for your pack chemistry.
- **TP4056 load sharing:** the cell charges while also powering the SIM800L; fine for a prototype, but a proper power-path charger is better for a product.
- **Credentials** are compiled in; keep the source private and give each tracker its own device user.